

Bank Customer Churn Prediction

ALY6015 Intermediate Analytics

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Project Overview & Research Questions



PROJECT GOAL



Identify factors associated with customer churn



Build a predictive model to classify high-risk customers



Support proactive customer retention strategies



DATA SET



Bank Customer Churn Dataset



10,000 customer records



14 variables



RESEARCH QUESTIONS

1

Which demographic and financial variables are the strongest predictors of customer churn?

2

Can a logistic regression model accurately classify high-risk customers?

3

Can churn be predicted early enough to support business decision-making?

4

How do geography, age and account balance influence churn probability



Methods Used



GOAL: Build and interpretable and effective model to predict customer churn



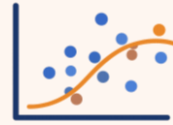
Data Cleaning & Preparation

- Checked missing values
- Review and correct data types
- Convert categorical variables to factors
- Create training and testing



Exploratory Data Analysis (EDA)

- Analyze variable distributions
- Examine churn patterns across demographics, location and activity
- Correlation analysis
- Scatterplot: Age vs Balance by churn status



Logistic Regression Modelling

- Build logistic regression model to predict churn (Exited)



Model Optimization

- **Threshold Tuning:** Identify optimal probability threshold (selected 0.30)
- **SMOTE (Class Balancing):** Address class imbalance between churn vs non-churn
- **VIF Diagnostics:** Check multicollinearity among predictors



Model Evaluation

- Confusion matrix
- Accuracy
- Recall (Sensitivity)
- Specificity
- Balanced Accuracy
- ROC Curve & AUC (Discriminatory Ability)

Components of model optimization



Threshold Tuning

Evaluate multiple thresholds to balance recall (detecting churners) and specificity (avoiding false alarms). Selected 0.30



SMOTE (Class Balancing)

Apply SMOTE on the training data to handle class imbalance and improve recall for churned customers

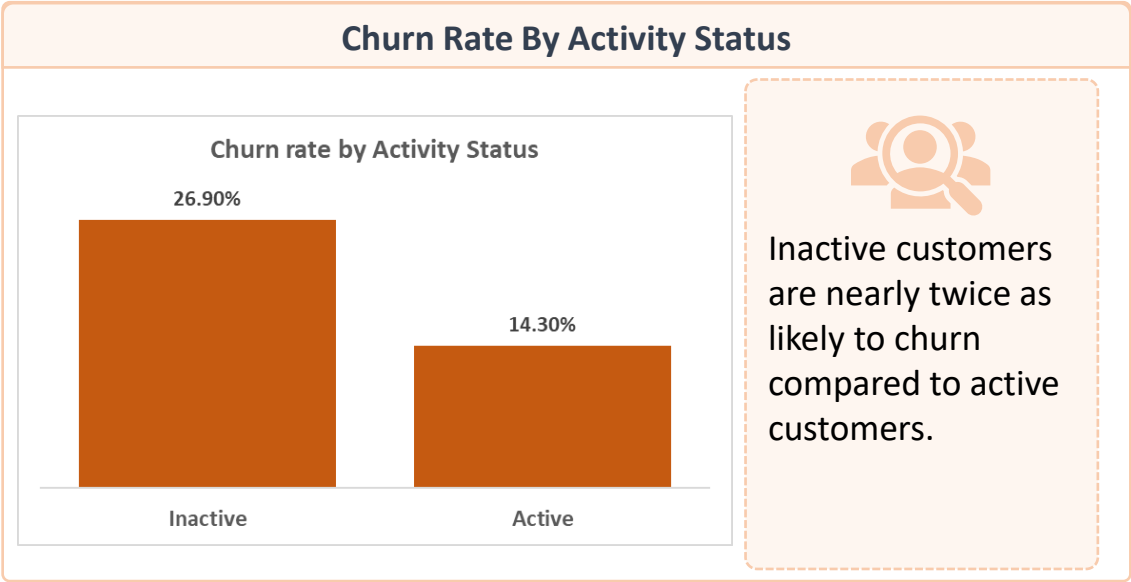
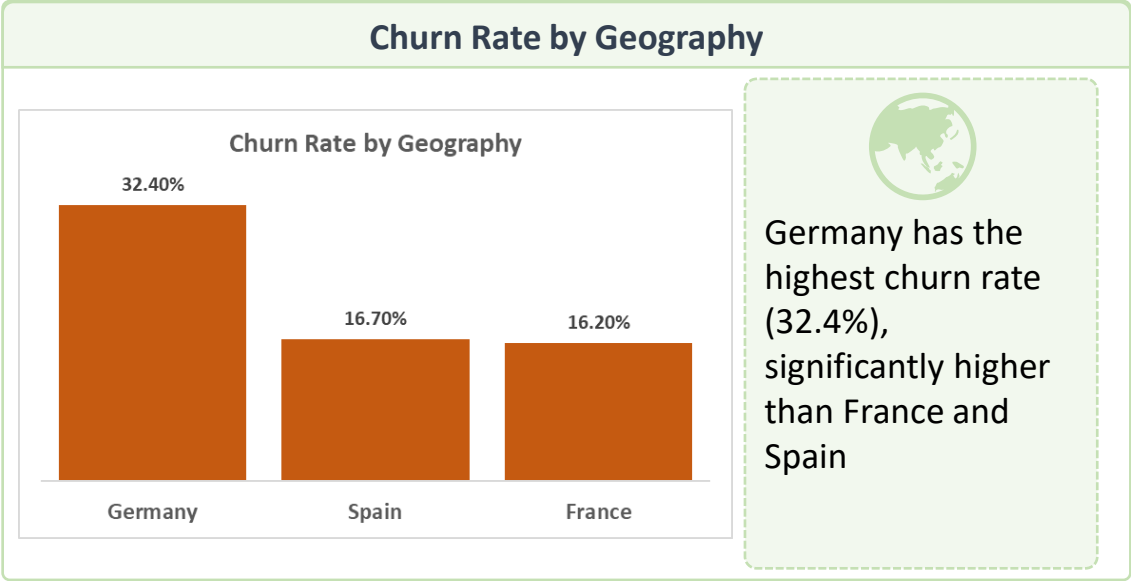
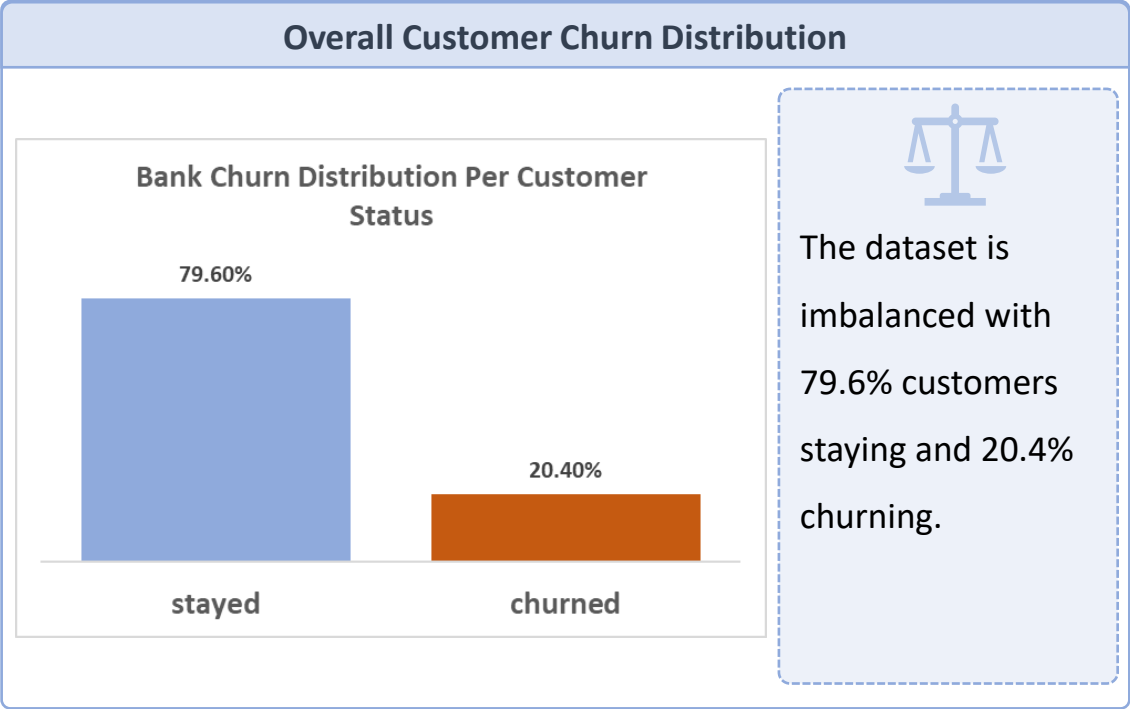


VIF Diagnostics

Check Variance Inflation Factor (VIF) to detect and address multicollinearity among predictors.

Exploratory Data Analysis (EDA)

Understanding customer churn patterns across overall distribution, geography and activity status.



Key Insights

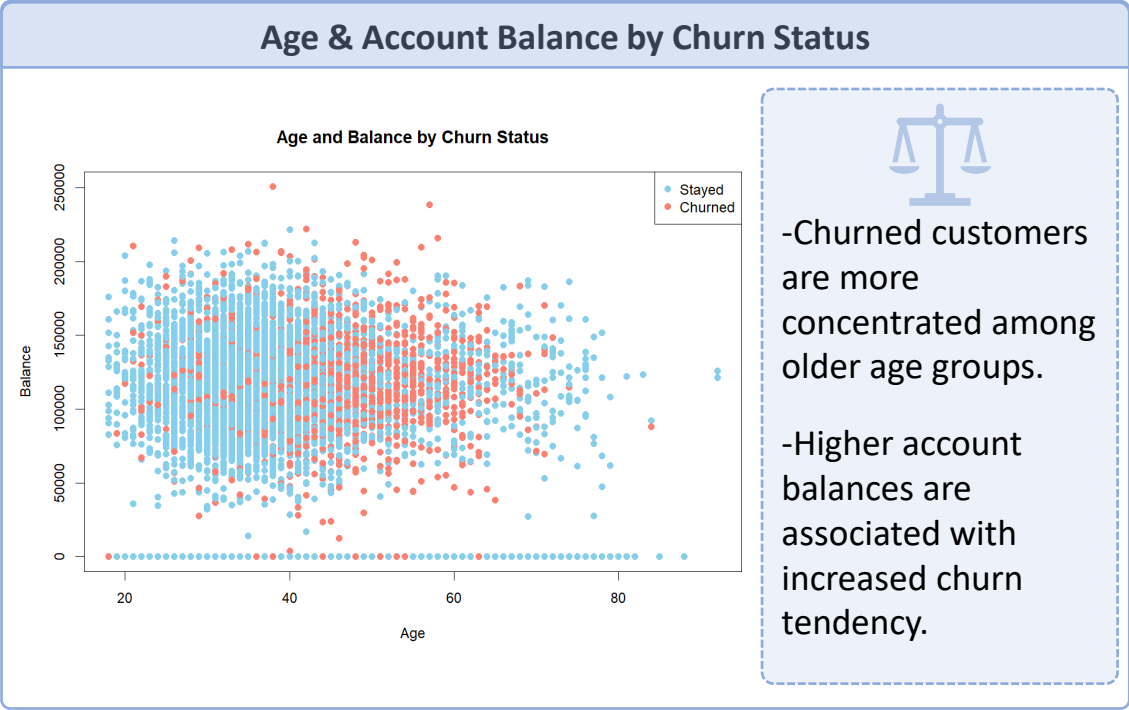
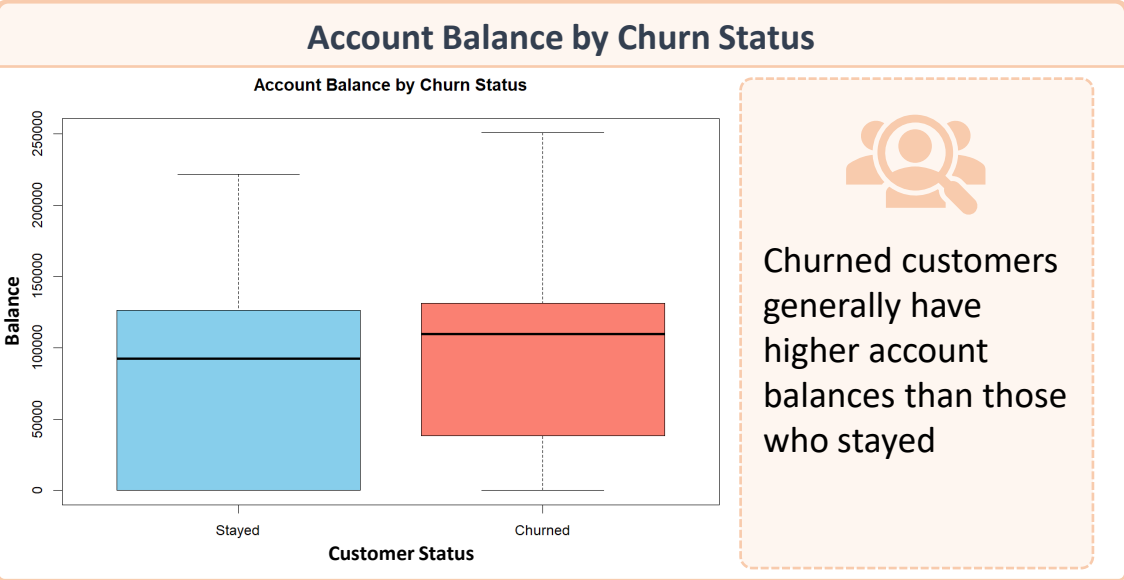
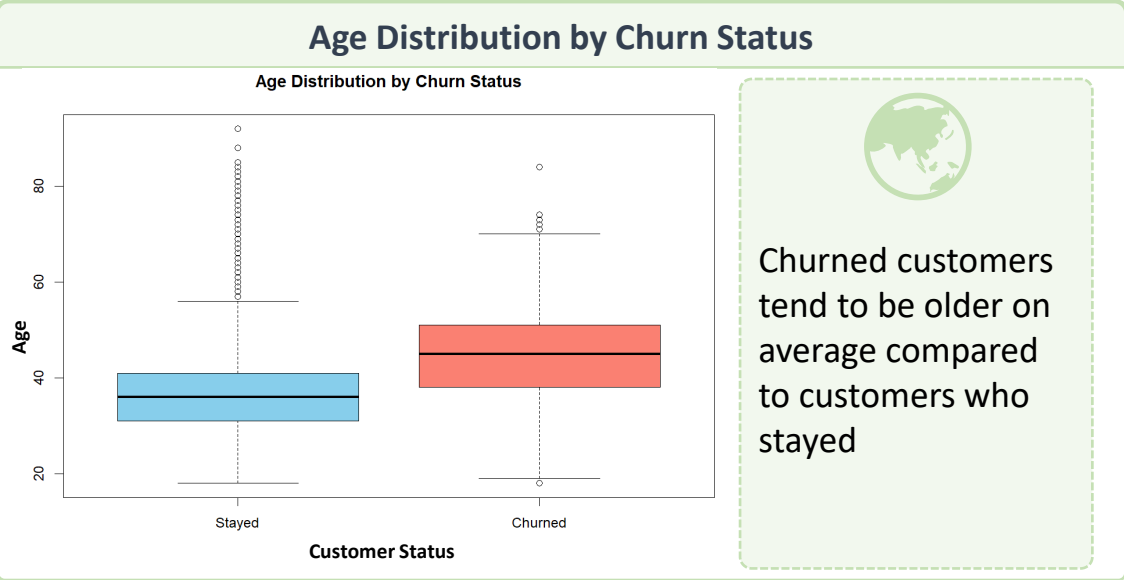
The dataset is **imbalanced**, which is important to consider during modeling and evaluation.

Germany shows the highest churn rate, indicating potential regional factors impacting customer retention.

Inactive customers have a much higher probability of churning.

Exploratory Data Analysis (EDA)

Understanding how age and account balance relate to churn behavior.



Key Insights



Older customers are more likely to churn.



Customers with higher account balances show higher churn tendency.



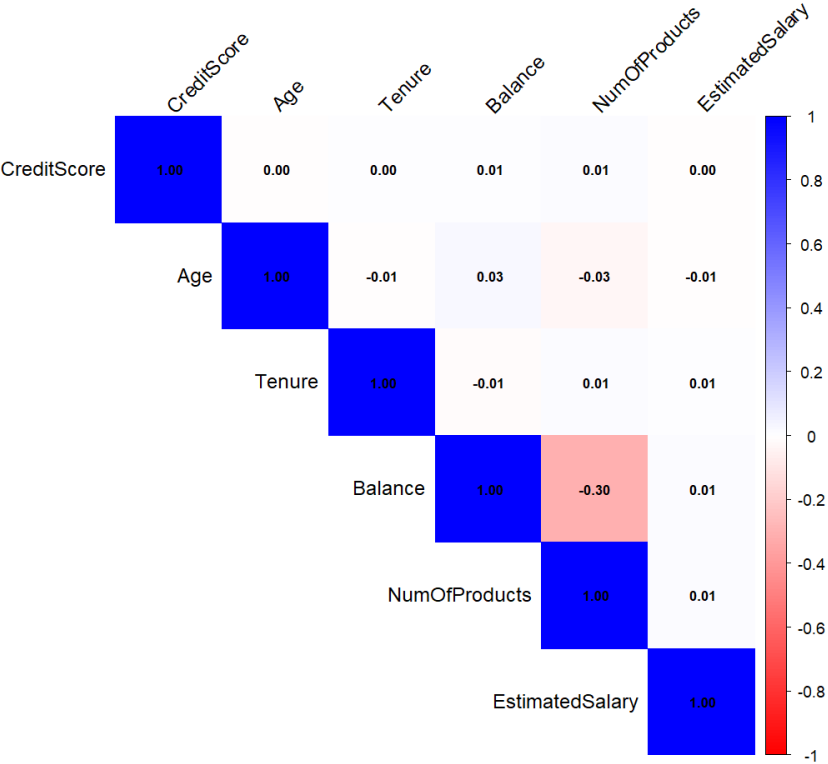
No strong linear relationship between age and balance, indicating other factors also play a key role.

Exploratory Data Analysis (EDA)

Correlation Analysis

We examined the relationships between the numeric variables to identify potential association and multicollinearity

Age Distribution by Churn Status



Key Insights



Most variables show **very weak correlations** with each other (correlation values close to 0).



Balance and NumOfProducts have a moderate negative correlation of -0.30.



No strong correlations (>0.7) were detected, indicating **low risk of multicollinearity** among predictors.



This supports the robustness of our logistic regression model.



Conclusion: The correlation analysis showed that most predictor variables had weak relationships with one another. This suggests a **low risk of multicollinearity** and **supports the suitability of logistic regression modeling**.

Logistic Regression Results

We built a logistic regression model to identify key drivers of customer churn.

Significant Predictors of Churn			
Predictor	Coefficient	p-value	Significant
CreditScore	-0.0007	0.031	Yes
GeographyGermany	0.801	<0.001	Yes
GeographySpain	0.064	0.447	No
GenderMale	-0.531	<0.001	Yes
Age	0.074	<0.001	Yes
Tenure	-0.02	0.081	No
Balance	0.000002	<0.001	Yes
NumOfProducts	-0.131	0.02	Yes
HasCrCard	-0.029	0.679	No
IsActiveMember	-1.068	<0.001	Yes
EstimatedSalary	0.000001	0.122	No

Significant predictors = (p value <0.05)

Odds Ratio (Interpretation)		
Predictor	Odds Ratio	Interpretation
CreditScore	0.999	Slight decrease in churn odds
GeographyGermany	2.227	Higher churn odds than France
GeographySpain	1.066	Similar churn odds to France
GenderMale	0.588	Lower churn odds than females
Age	1.076	Higher churn odds with increasing age
Tenure	0.981	Slight decrease in churn odds
Balance	1	Minimal positive effect
NumOfProducts	0.877	Lower churn odds with more products
HasCrCard	0.971	Minimal effect
IsActiveMember	0.344	Substantially lower churn odds
EstimatedSalary	1	Minimal effect

Key Insights	
• Germany customers were significantly more likely to churn than customers in France	• <i>OR = 2.227, p < 0.001</i>
	• Older customers showed higher churn likelihood
• Active customers had substantially lower churn risk	• <i>OR = 1.076, p < 0.001</i>
	• <i>OR = 0.344, p < 0.001</i>
• Customers with more products were less likely to churn	• <i>OR = 0.877, p = 0.020</i>
	• Account balance showed a statistically significant but very small positive effect on churn probability
• Credit score had only a minimal effect on churn	• <i>OR = 1, p < 0.001</i>
	• <i>OR = 0.999, p = 0.031</i>
• GeographySpain, HasCrCard, Tenure, and EstimatedSalary were not statistically significant predictors	• <i>p-values > 0.05</i>



Conclusion: Geography, age, customer activity status, and account balance emerged as the strongest drivers of customer churn. Germany customers and older customers showed significantly higher churn likelihood. Active customers and customers with more products were less likely to leave the bank.

Multicollinearity Diagnostics (VIF)

Variance Inflation Factor (VIF) was used to assess multicollinearity among predictors

Variance Inflation Factor (VIF) Results

Predictor	Adjusted VIF
CreditScore	1.001
Geography	1.05
Gender	1.002
Age	1.04
Tenure	1.002
Balance	1.133
NumOfProducts	1.037
HasCrCard	1.002
IsActiveMember	1.039
EstimatedSalary	1.001

Key Insights

- All adjusted VIF values were close to 1.
- No evidence of multicollinearity was detected.
- Predictor variables were considered stable and reliable for logistic regression modeling.

Model Evaluation and Optimization

We evaluated different modeling strategies to improve churn detection performance.

Model Optimization Strategies

1. Baseline Model
(Threshold = 0.50)



Accuracy

82.2%

Recall

23.1%



Missed many churned customers

Metric	Value
Accuracy	82.16%
Sensitivity (Recall)	23.08%
Specificity	97.28%
Precision	61.50%
Kappa	0.24

2. Threshold-Tuned Model
(Threshold = 0.30)



Accuracy

79.2%

Recall

51.2%



Best balance between churn detection and model stability

Metric	Value
Accuracy	79.23%
Sensitivity (Recall)	51.88%
Specificity	86.22%
Precision	45.60%
Kappa	0.37

3. SMOTE Model
(Threshold = 0.30)



Accuracy

58.9%

Recall

86.1%



Recall improved to 82.9%

Produced excessive false +

Lower overall accuracy and specificity

Metric	Value
Accuracy	58.95%
Sensitivity (Recall)	86.09%
Specificity	52.01%
Precision	30.50%
Kappa	0.15

Key Insights



Lowering the threshold to 0.30 improved recall while maintaining reasonable accuracy



SMOTE significantly increased recall but at the cost of more false positives and lower overall performance



The threshold-tuned model (0.30) provides the best balance for churn detection.

Model Comparison Summary				
Model	Accuracy	Sensitivity (Recall)	Specificity	Kappa
Baseline (Threshold = 0.50)	82.16%	23.08%	97.28%	0.24
Threshold-Tuned (Threshold = 0.30)	79.23%	51.88%	86.22%	0.37
SMOTE + Threshold = 0.30	58.95%	86.09%	52.01%	0.15

Class distribution Before and After SMOTE		
Class	Before SMOTE	After SMOTE
Stayed (0)	79.60%	56.60%
Churned (1)	20.40%	43.40%

Model Comparison Summary

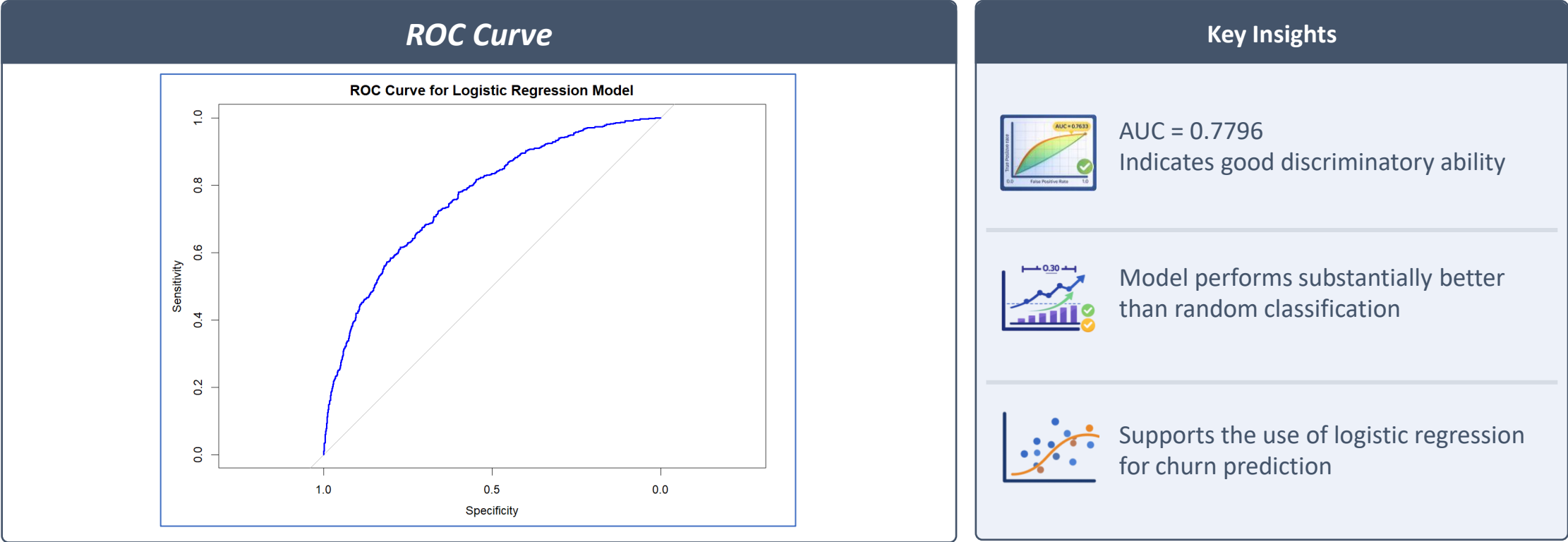
	Baseline Model (Threshold = 0.50)	Threshold-Tuned Model (Threshold = 0.30)	SMOTE + Threshold (Threshold = 0.30)
 Catches churners (recall)	23.08% Missed many customers who actually churned	51.88% Catches more than double the churners vs. baseline.	86.09% Catches the most churners
 Correctly identified loyal customers (Specificity)	97.28% Very good at identifying loyal customers	86.22% Still good at identifying loyal customers	52.01% Many loyal customers incorrectly flagged as high risk.
 Overall Correct Prediction (Accuracy)	82.16% 82 out of 100 predictions were correct	79.23% 79 out of 100 predictions were correct	58.95% Only 59 out of 100 predictions were correct.
 Overall Balance (Kappa)	0.24 Lower overall balance	0.37 Best overall balance among all models.	0.15 Lowest overall balance
 Business impact (Cost)	Lowest operational cost, but high risk of missing churners	Moderate operational cost with strong improvement in churn detection.	Highest operational cost due to many false alarms and unnecessary interventions.



The threshold-tuned model (0.30) provides the best balance between detecting churners and keeping false alarms at a reasonable level, making it the most practical and cost-effective choice for the bank.

ROC Curve and AUC

The ROC Curve and AUC were used to evaluate the overall discriminatory ability of the logistic regression model



Final Selected Model



Threshold-tuned Logistic Regression Model (0.30)



Reason:

- Most balanced predictive performance
- Better churn detection without excessive false positives

Business Recommendations

Actionable recommendations to reduce customer churn and strengthen long-term customer relationships



1

Prioritize retention efforts for inactive customers
Send personalized re-engagement offers and account activity reminders.



2

Focus retention strategies on Germany
Launch targeted loyalty campaigns for Germany-based customers.



3

Monitor older customers with high balances
Assign relationship managers and provide personalized financial support.



4

Implement proactive customer engagement programs
Conduct regular customer check-ins and satisfaction outreach.



5

Use predictive analytics for early churn intervention
Automatically flag high-risk customers for retention follow-up.



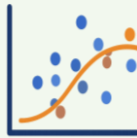
Early identification of high-risk customers

Detect potential churners early and take timely action to retain them



More targeted retention campaign

Focus resources on the right customers with personalized and relevant offers



Improve customer relationship management

Stronger relationships lead to higher satisfaction, loyalty, and long-term value.



By leveraging predictive insights, the bank can take proactive actions to reduce churn, strengthen customer loyalty, and drive long-term business growth.

THANK YOU

